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CALCULATION OF X4 AND PLOTTING OF 3D GRAPHS FOR VEU

The article below discusses the issue of calculating the variable X4, which affects the values of GDP (GDP). The values obtained during the calculations are shown in the form of 3D graphs for X4. In this case, the variables are constant, decrease and increase. That is, the article considers the dependence of the change $X4 = f(X1, X2, X3, X5, X6)$ [1, 2, 3].

Here, Figures 1 and 2 show 3D graphs for X4 with $X1= X2= 10..1$, $X3= 1$, $X5= 1..10$, $X6= 9.99..1$ and $X1= 10..1$, $X2= X3= 1$, $X5=1..10$, $X6= 9.99..1$. So in Figure 1, the value of X4 has a minimum of 0.86909 at point 6, and in Figure 2 it decreases by 1.27 times.

The following two Figures 3 and 4 show 3D graphs for X4, when the variables were $X1= X6= 10..1$, $X2= X5=1..10$, $X4=1$ and $X1= X5= 1..10$, $X2= X6= 10..1$, $X4= 1$ respectively. Here in Figures 3 and 4, the X4 values have minima of 0.99978757 and 0.99978757 at points 6.

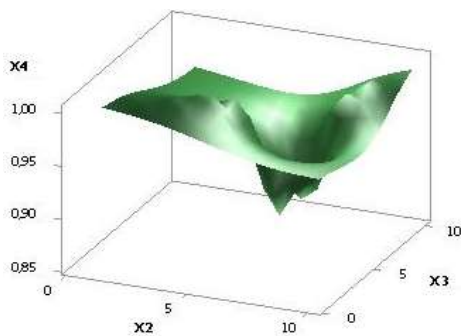


Fig.1. $X4 = f(X1, X2, X3, X5, X6)$

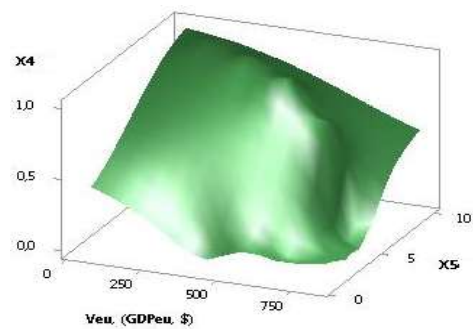
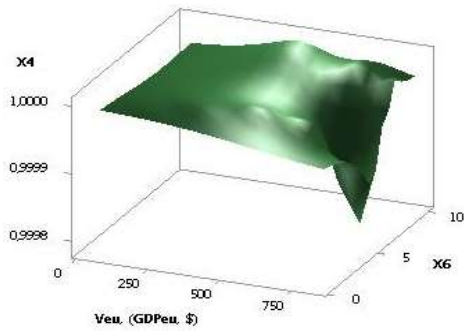
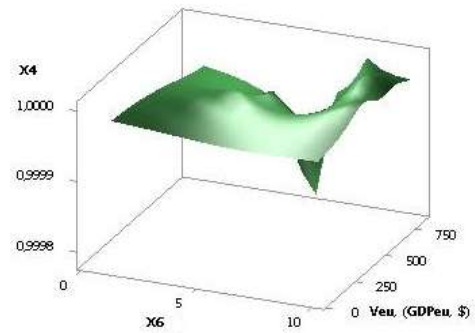
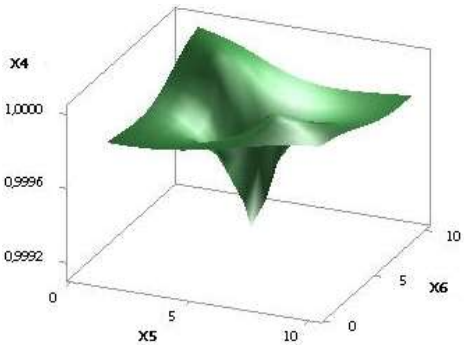
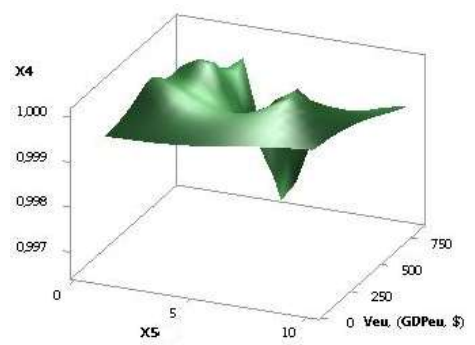
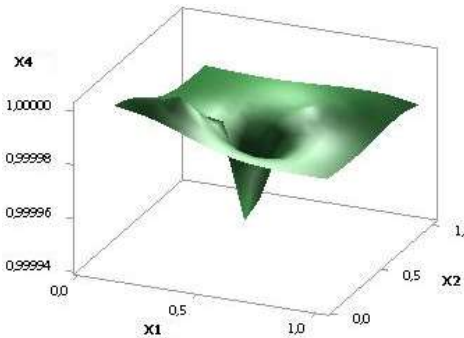
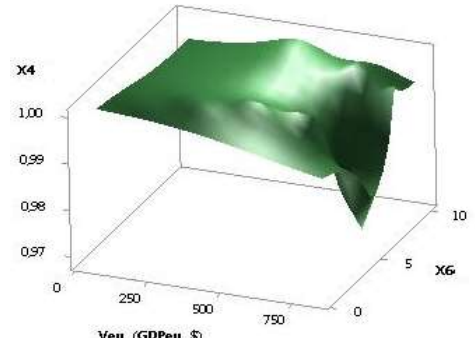


Fig.2. $X4 = f(X1, X2, X3, X5, X6)$

The following two figures 5 and 6 were constructed at $X1= X3= 1..01$, $X2= X5=1..10$, $X6= 1..9.99$ and $X1= X5=1..10$, $X2= X3=1..01$, $X6= 1..9.99$. Here in Figures 5 and 6, the X4 values also have minima of 0.99915 and 0.9966 at points 6.


Fig.3. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

Fig.4. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

Fig.5. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

Fig.6. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

To construct two 3D graphs in Figures 7 and 8, the following values of the variables $X_1 = X_2 = X_5 = 1..10$, $X_3 = 1..01$, $X_6 = 1..9.99$ and $X_1 = X_2 = X_3 = 1..01$, $X_5 = 1..10$, $X_6 = 1..9.99$ were used. Here in Figures 7 and 8, the X_4 values also have minima of 0.9999468979 and 0.96893 at points 6.


Fig.7. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

Fig.8. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

The following figures 9 and 10 were constructed at $X_1 = 1$, $X_2 = 10..1$, $X_3 = 1..01$, $X_5 = 1..10$, $X_6 = 1..9.99$ and $X_1 = X_2 = 10..1$, $X_3 = 1..01$, $X_5 = 1..10$, $X_6 = 1..9.99$. Here in Figure 9, the values of X_4 decrease by 1.06 times, and in Figure 10 it has a minimum of 0.96893 at point 6. The last figures 11 and 12 were constructed at $X_1 = 10..1$, $X_2 = X_3 = 1..01$, $X_5 = X_6 = 1..10$ and $X_1 = X_3 = 1..01$, $X_2 = 10..1$, $X_5 = X_6 = 1..10$. Here in Figure 11, the values of X_4

decrease slightly by 1.27 times, and in Figure 12 they have a minimum of 0.9998641 at point 7.

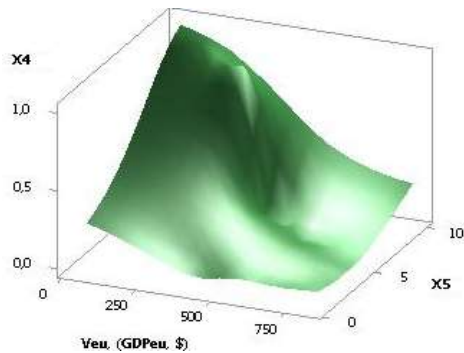


Fig.9. $X4 = f(X1, X2, X3, X5, X6)$

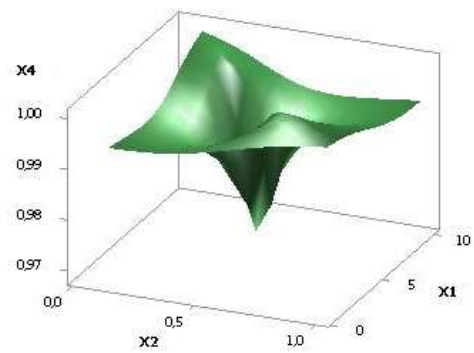


Fig.10. $X4 = f(X1, X2, X3, X5, X6)$

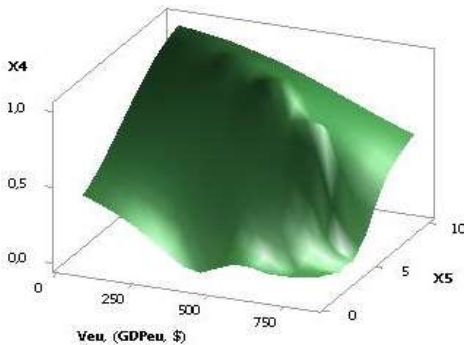


Fig.11. $X4 = f(X1, X2, X3, X5, X6)$

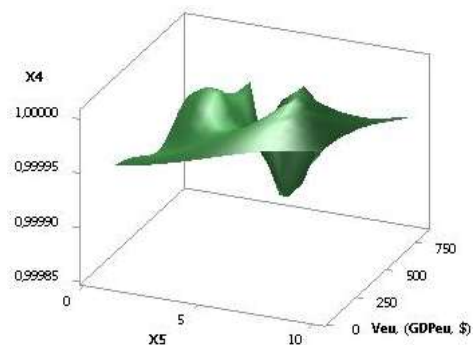


Fig.12. $X4 = f(X1, X2, X3, X5, X6)$

REFERENCES

1. Pil E.A. Analysis of 2D Veu graphs using the X4 variable // FUNDAMENTAL AND APPLIED SCIENTIFIC RESEARCH: CURRENT ISSUES OF ACHIEVEMENTS AND INNOVATIONS: Part 1. Collection of articles of the International Scientific and Practical Conference (August 12, 2021, Kazan). / at 2 h. h.1. / -Ufa: OMEGA SCIENCE, 2021 - 270 p. – pp. 227-229
2. Pil E.A. Variants of 2D graphs of Veu when calculating the variable X4 // FUNDAMENTAL AND APPLIED SCIENTIFIC RESEARCH: TOPICAL ISSUES OF ACHIEVEMENTS AND INNOVATIONS: Part 1. Collection of articles of the International Scientific and Practical Conference (August 12, 2021, Kazan). / at 2 h. h.1. / -Ufa: OMEGA SCIENCE, 2021 - 270 p. – pp. 229-231
3. Pil E.A. Calculation of X4 in the construction of 2D graphs for Veu // FUNDAMENTAL AND APPLIED SCIENTIFIC RESEARCH: TOPICAL ISSUES OF ACHIEVEMENTS AND INNOVATIONS: Part 1. Collection of articles of the International Scientific and Practical Conference (August 12, 2021, Kazan). / at 2 h. h.1. / -Ufa: OMEGA SCIENCE, 2021 - 270 p. – pp. 231-234